

SHRAVAN KANNEPALLI

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EDUCATION

Bachelor of Science in Mechanical Engineering

May 2028

University of Maryland, College Park

2024 & 2025 Fall Semester Dean's List

WORK EXPERIENCE

Geospatial Analysis Intern, NASA

June 2023 - August 2023

- Engineered a geospatial database by scraping and analyzing AWS satellite data, enabling the processing of complex queries to monitor global environmental shifts.
- Developed a real-time tracking system for significant Earth events, providing the scientific community with automated updates and high-fidelity data insights.

PROJECTS

Over-Terrain Autonomous Vehicle

September 2024 - December 2024

- Collaborated within a five-person team to design an autonomous Over-Terrain Vehicle, earning a Top 5 Innovative Project Nomination.
- Integrated a custom collection system featuring a puck-and-aligner mechanism, pump, limit switch, and conductivity sensor, interfaced via Arduino to automate water sampling.
- Served as Budget Analyst, for the raw materials and electrical components to ensure the project remained within strict financial constraints.
- Contributed to vehicle construction, assisting in the physical assembly and structural optimization of the chassis to navigate varied terrain.

Structural Truss Design & Analysis

March 2025 - May 2025

- Designed and fabricated a structural truss utilizing oak members and high-strength adhesive joints in a team, engineered to safely support specific load requirements.
- Conducted structural analysis, calculating support reactions and evaluating normal, shear, and buckling stresses to ensure mechanical integrity.
- Calculated safety factors for all critical components, validating the design against theoretical failure limits and material properties.
- Optimized member dimensions and layout to accurately predict failure modes and maximize the strength-to-weight ratio.

L1 Rocket Certification

October 2025 - November 2025

- Earned L1 Rocket Certification by designing and successfully launching a high-power model rocket within an accelerated development cycle.
- Utilized OpenRocket simulation software to model complex flight dynamics and center of pressure/gravity relationships to validate aerodynamic stability.
- Engineered recovery systems and verified flight profiles through digital simulations to ensure safe vehicle descent and hardware reuse.

Beam Design Project

October 2025 - December 2025

- Designed a pine built-up I-beam with a team, engineered to maximize strength-to-weight ratio by balancing failure margins across multiple structural modes.
- Performed high-capacity three-point bending tests to evaluate uniform performance and observe real-world material behavior under significant loading.
- Analyzed critical fracture points and failure mechanisms to validate the engineering approach and refine the structural efficiency of the design.

ACTIVITIES

DaCadence A Cappella | Vice President

- Assist in leading and organizing a university a cappella group, coordinating rehearsals and performances.

Terrapin Rocket Club (Aerostructures/Two Stage)

- Active member contributing the Aerostructures Team for the International Rocket Engineering Competition (IREC) and the Two-Stage Rocket Team. Collaborate with team members on design, fabrication, and integration challenges to support the development and successful launch of competition rockets.
- Currently working on failure analysis and building of the main/drogue parachute for next years competition rocket

FLL Mentoring

- Mentored a FIRST LEGO League (FLL) team by guiding students through the robot design cycle and a specialized research project.
- Facilitated technical growth by teaching programming logic, iterative testing, and fundamental engineering principles.
- Instilled core design values and professional ethics to help the team translate creative concepts into functional, competition-ready solutions.

SKILLS

CAD: Solidworks/ Fusion, Circuit Design & Soldering, Coding: (Python, Java, C++, Matlab), Mentoring/Teaching